

Quarter I review paper · Algebra Grade 11

40 minutes 100 marks 13 questions + bonus No calculator

Lessons 5–22, and limits at infinity

- Answer on separate paper, and show your working — method carries marks.
- State the nature of every stationary point, and the test you used.

NAME _____ CLASS _____ DATE _____ MARK _____ / 100

A Section A · Five marks each

EASY

30 marks

Short answers — one or two lines each.

1 Differentiate: $y = 4x^3 - 5x + 7$ [5]

2 Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$ [5]

3 Differentiate: $y = (3x + 1)^5$ [5]

4 Solve: $|2x - 6| = 10$ [5]

5 Find the stationary point of $y = x^2 - 6x + 5$ and state its nature. [5]

6 Solve: $|x - 3| < 5$ [5]

B Section B · Eight marks each

MEDIUM

40 marks

Method marks are available even when the final answer is wrong.

7 Evaluate, showing the division in each case:

(a) $\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 1}{5x^2 + 4}$ [3]

(b) $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 - 2}$ [2]

(c) Explain why $\lim_{x \rightarrow \infty} \frac{x^3 + 2x}{4x^2 - 1}$ does not exist. [3]

8 $f(x) = x^2 - 3x$.

(a) Show that $\frac{f(2+h) - f(2)}{h} = h + 1$. [5]

(b) Hence write down $f'(2)$. [3]

9 Differentiate $y = \frac{2x - 1}{x^2 + 3}$ and find the gradient of the curve at $x = 1$. [8]

- 10 The curve $y = x^3 - 4x$. Find the equation of the tangent and the equation of the normal at the point where $x = 2$. [8]
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- 11 $y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ and its value at $x = 4$. [8]
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C Section C • Fifteen marks each

Full method required.

HARD

30 marks

- 12 $y = x^3 - 6x^2 + 9x$.
- (a) Find the stationary points and determine the nature of each. [6]
 - (b) State the intervals on which the function is increasing and decreasing, and sketch the curve. [5]
 - (c) Find the values of k for which $x^3 - 6x^2 + 9x = k$ has three distinct solutions. [4]
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- 13 The curve $y = \sqrt{2x + 7}$.
- (a) Find $\frac{dy}{dx}$. [4]
 - (b) Find the equation of the tangent at the point where $x = 1$. [6]
 - (c) Find the equation of the normal at that point, and the coordinates of the point where it crosses the x -axis. [5]
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Bonus

- ★ Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100.

BONUS

+10 marks

- 14 Find the equations of the two tangents to $y = x^2$ that pass through the point $(0, -4)$.
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